# Brightening Low-Light Images Using Diffusion Models with Plug-and-Play Capability

**ECE 570: Artificial Intelligence**
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**Abstract**

Image restoration aims to recover high-quality images from degraded inputs affected by noise, blur, or poor illumination. While diffusion models have shown strong performance in denoising and deblurring, this work explores their extension to illumination enhancement. This project builds upon Zhu et al.'s "Denoising Diffusion Models for Plug-and-Play Image Restoration" by adapting the framework to brighten low-light images. A conditional diffusion model is trained using paired dark and bright facial images, enabling reconstruction of well-lit outputs from dark inputs. The training pipeline is adapted from OpenAI's guided-diffusion framework with custom modifications for image-to-image translation. A new inference function is implemented to support batch processing and efficient deployment. Experimental results demonstrate improved visibility in low-light scenarios, though some outputs still exhibit residual noise and color shifts. These findings highlight the flexibility of diffusion-based generative models for restoration tasks and their potential in illumination enhancement applications.

*The source code is available at:*
https://github.com/MisterUnknownGuyPerson/Project570.git
*The demo video is available at (also on git):*
https://drive.google.com/file/d/1HZLTIKme87oBS-yHr0nJv7Yj-MMDJUm8/view?usp=drive_link

## 1. Introduction

Image restoration encompasses a range of techniques aimed at recovering high-quality images from degraded versions affected by noise, blur, or low resolution. In recent years, diffusion probabilistic models have demonstrated state-of-the-art performance across a variety of restoration tasks, such as deblurring and denoising. These models utilize a learned reverse diffusion process to progressively reconstruct clean signals from noise, offering both flexibility and high fidelity in image generation. This project extends the capabilities of diffusion-based image restoration to a less-explored degradation class: insufficient illumination.

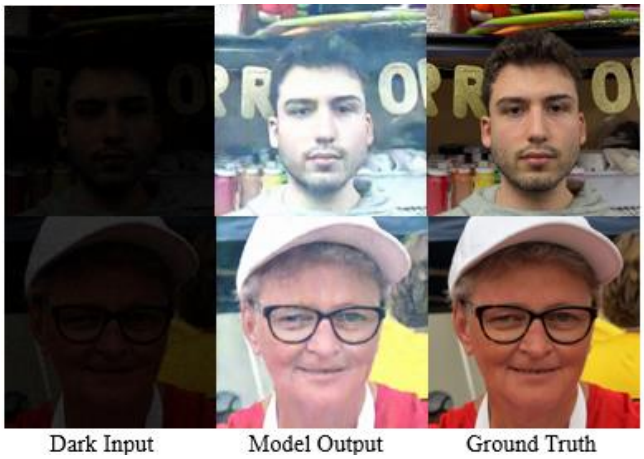


*Figure 1:* Examples of reconstructions. Presenting the reconstructed images including corresponding labels. "Dark Input" is synthetically generated from the ground truth.

The primary objective is to enhance low-light images by training a conditional diffusion model to reconstruct well-lit versions from dark inputs. This work builds on the plug-and-play framework introduced by Zhu et al. [5], extending its application to the task of illumination enhancement. To enable image-to-image translation, the model is conditioned on dark input images and trained using paired examples from the FFHQ dataset [4]. Dark images were synthetically generated through controlled Python-based transformations, including brightness reduction, Gaussian noise injection, and gamma adjustment. This synthetic pairing ensured pixel alignment between the dark and bright versions, which is essential for effective conditioning during training.

The training process adapts OpenAI's guided-diffusion framework [1], originally designed for class-conditional generation, to support conditional image translation. Several implementation changes were made to accommodate this new setting. These include modifying the model to accept concatenated dark-bright input channels, creating a new training script for paired data, and developing a new inference function that enables batch processing and efficient deployment across directories.

The results align with the goal and show that the trained model successfully enhances visibility from low-light conditions (nearly full black images), producing brighter and sharper outputs. However, some outputs exhibit artifacts, such as residual noise or dominant color bias, indicating areas for future refinement. This demonstrates that diffusion models can be effectively extended to tasks such as low-light enhancement, paving the way for future work on robust generative restoration systems. Such enhancements hold significance for practical deployment in applications like CCTV visibility restoration, night-vision hardware, or mobile photography, where software-based solutions can complement or even replace physical sensors in poor lighting conditions. Figure I presents the results from this model and compares them to the ground truth.

## 2. Related Work

Image restoration has seen significant advancements with the emergence of diffusion-based generative models. Among these, Zhu et al.'s framework introduces a plug-and-play approach that adapts denoising diffusion probabilistic models (DDPMs) for restoration tasks [5]. Rather than training separate models for each degradation type (e.g., blur, noise, low resolution), their method relies on pre-trained diffusion models that are guided to iteratively reconstruct clean images from degraded observations.

The restoration process is formulated using the Half-Quadratic Splitting (HQS) optimization technique, which separates the data consistency term from the learned prior term [5]. In practice, this enables alternating updates: one step applies Gaussian denoising using a pre-trained diffusion model, while the other enforces consistency with the degraded input based on the known degradation operator. This flexible formulation makes the model applicable across a wide range of tasks, including denoising, deblurring, and super-resolution. HQS algorithm:

$$\begin{cases} \mathbf{z}_k = \arg\min_{\mathbf{z}} \frac{1}{2(\sqrt{\lambda/\mu})^2} \|\mathbf{z} - \mathbf{x}_k\|^2 + \mathcal{P}(\mathbf{z}) & \text{(1a)} \\ \mathbf{x}_{k-1} = \arg\min_{\mathbf{x}} \|\mathbf{y} - \mathcal{H}(\mathbf{x})\|^2 + \mu\sigma_n^2 \|\mathbf{x} - \mathbf{z}_k\|^2 & \text{(1b)} \end{cases}$$

where the parameter $\boldsymbol{\mu}$ is introduced as the coefficient for the data-consistent constraint term. Here the subproblem (1a) with prior term is a Gaussian denoising problem, and the subproblem (1b) with the data term is a proximal operator [6] which usually has a closed-form solution that depends on $\mathcal{H}$.

An important innovation in Zhu et al.'s method is the use of task-specific conditioning that directs the reverse diffusion process. This conditioning allows the diffusion model to be used as a powerful image prior, even when the observed data is not purely Gaussian noise. Unlike unconditional generation, the process starts from the degraded image and refines it through diffusion, improving convergence and enabling plug-and-play generalization.

This project highlights a novel extension of Zhu et al.'s framework by adapting it to a new degradation domain: image illumination. While the original work focused on general restoration tasks like deblurring and denoising, this implementation differs from prior work by targeting low-light enhancement. This demonstrates that plug-and-play diffusion can generalize visibility restoration under challenging lighting conditions. Furthermore, it builds upon Zhu et al.'s plug-and-play formulation by introducing a new conditional model trained from scratch using paired bright and dark facial images derived from the FFHQ dataset. Synthetic dark inputs are created via controlled transformations to preserve structural similarity while simulating varied illumination degradation. The training is conducted using a new script built on OpenAI's guided-diffusion framework [1], with architectural modifications to support six-channel conditioning. Lastly, a new inference pipeline is implemented to enable batch processing, allowing efficient deployment and experimentation across large-scale image datasets. This effectively enhances Zhu et al.'s work and contributes by extending its functionality to an extra image restoration task: image brightening.

## 3. Problem Definition

Image restoration involves reconstructing a clean image $x$ from a degraded observation $y$, where $\mathcal{H}$ denotes the degradation operator and $n \sim \mathcal{N}(0, \sigma_n^2)$ represents noise [5]. This is modeled as:

$$y = \mathcal{H}(x) + n \qquad (2)$$

This project focuses on a specific form of degradation: low-light illumination. Poor illumination can significantly reduce image quality, hindering visibility. An image brightening software could help create new night vision technologies or aid forensics by enhancing CCTV recordings. The objective is to enhance visibility in poorly lit images by learning a mapping from dark inputs $y$ to corresponding well-lit outputs $x$. To achieve this a data driven approach is used: training a conditional diffusion model to model the posterior $p(x|y)$ where $x$ is the desired bright image and $y$ is the observed dark image. Specifically, the forward process adds noise progressively to the bright image, while the learned reverse process reconstructs the bright image conditioned on the dark input.

Formally, the diffusion process defines a sequence of latent variables $x_0, x_1, \dots, x_T$ where $x_T \sim \mathcal{N}(0, I)$ and the model learns to sample $x_0$ (i.e., the bright image) conditioned on $y$ [2]. The inference task then becomes:

$$x_0 \sim p_\theta(x_0|y) = \int p_\theta(x_0|x_1, y) \dots p_\theta(x_{T-1}|x_T, y)\, dx \qquad (3)$$

In summary this project seeks to extend Zhu et al.'s plug-and-play diffusion model framework to the task of night vision. This is done by training a new model plug in play using the above detail logic and OpenAI's guided diffusion pipeline. Furthermore, a new inference function is created to enable batch processing and efficient deployment, facilitating practical restoration of night-time or poorly illuminated imagery.

## 4. Methodology

This work investigates the use of diffusion models for illumination enhancement in low-light facial images. A conditional diffusion model is trained using paired bright and synthetically darkened versions of the same image. The training pipeline is based on the OpenAI guided-diffusion framework [1], with substantial modifications to support image-to-image translation, while the inference pipeline builds on Zhu et al.'s code to enable plug-and-play deployment. In earlier experiments, DDIM sampling [3] was also explored to accelerate inference by reducing the number of sampling steps.

### 4.1 Dataset and Preprocessing

The dataset used for training is derived from the FFHQ face dataset, which consists of 1024×1024 high-quality human face images. All images were resized to 256×256 pixels to match the model input resolution. For each bright image $x$, a dark counterpart $y$ was generated using a Python-based preprocessing toolsets. This transformation involves applying:

- **Brightness reduction**: Subtractive darkening with randomized factors to ensure variation.
- **Gaussian noise addition**: Simulates sensor or environmental noise with variance scaled to local brightness.
- **Clipping and normalization**: Ensures the resulting images are valid RGB tensors in [0,1].

This results in a paired dataset of 10,000 dark/bright image pairs (20,000 images total). Preprocessing was performed in Google Colab, with data loaded directly from Google Drive.

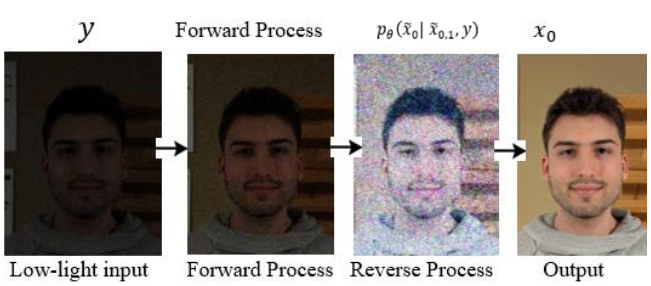


*Figure 2:* Process of image illumination using conditional diffusion model

### 4.2 Model Architecture and Training

The model architecture is adapted from OpenAI's guided-diffusion pipeline, which was originally designed for class-conditional or unconditional image generation. To enable image-to-image translation for illumination enhancement, the model is modified to operate under image-conditional guidance. Specifically, the input channel dimension is increased from 3 to 6 to allow concatenation of the dark input image $y$ with the noisy latent variable $x_t$ at each timestep. This conditioning mechanism is implemented by adjusting the input layer of the U-Net backbone to accept six-channel tensors. It enables the model to leverage both the degraded observation and its latent representation during the reverse diffusion process, effectively learning the transformation from low light to well-lit images by modeling the conditional posterior $p_\theta(x_0 | y)$.

A custom training script is developed to handle the conditioning pipeline, supporting the modified data loader and tensor structure. The model is trained on paired image datasets consisting of bright facial images and their artificially darkened counterparts. Multiple separate training regimes were tested, with notable regimes being: 1,000 image pairs with 10,000 iterations and 10,000 with 20,000 iterations.

Training was performed on Google Colab Pro using the NVIDIA A100 GPU, with the FFHQ dataset and augmentation scripts mounted from Google Drive. This configuration allowed for scalable training and rapid experimentation within a cloud-based environment.

**Algorithm 1** above provides a high-level description of the training process.

Algorithm 1: Training Conditional Diffusion Model for Illumination Enhancement

Input: Dataset of paired images ( $y_i, x_i$ ), model $f_\theta$ , total steps T
Output: Trained conditional model $f_\theta$
for each training step do
  Sample a pair $(y, x)$ from dataset
  Sample timestep $t \sim Uniform\ (1, T)$
  Compute noisy image $x_t \leftarrow \sqrt{\bar{a}_t} x + \sqrt{1 - \bar{a}_t}\ \varepsilon$, where $\varepsilon \sim \mathcal{N}(0, I)$
  Form conditional input: input $\leftarrow [x_t, y] \in \mathbb{R}$^$(6 \times H \times W)$
  Predict noise: $\varepsilon_\theta \leftarrow f_\theta$(input, t)
  Compute loss: L $\leftarrow ||\varepsilon - \varepsilon_\theta||^2$
  Backpropagate and update $f_\theta$
end for

Algorithm 2: Inference for Image Brightening via Conditional Diffusion

Trained model $f_\theta$, dark input image $y$, diffusion steps $T$, conditioning flag

Output: Brightened image $\hat{x}$

Initialize $x_T \sim \mathcal{N}(0, I)$
for $t = T$ $to$ 1 do:
    Concatenate $x_t$ and $y \rightarrow input \leftarrow [x_t, y]$
    Predict noise $\varepsilon_\theta \leftarrow f_\theta(input, t)$
    Compute $x_{t-1}$ using DDIM
end for
Return final estimate $\hat{x} \leftarrow x_0$

### 4.3 Inference and Evaluation

Once trained, the model is deployed without further task-specific retraining, consistent with the plug-and-play paradigm proposed in Zhu et al.'s work. To support practical deployment, a custom Python inference pipeline was implemented using PyTorch and executed locally on a workstation equipped with an NVIDIA RTX 3060 GPU and 16 GB of RAM. The script supports batch inference from a directory of dark images, enabling scalable evaluation across diverse test inputs.

Inference was performed using varying numbers of reverse diffusion steps, specifically 25, 100, and 1000, to investigate the trade-off between computational efficiency and reconstruction quality. The "ddim25" setting, derived from the DDIM sampling method introduced by Song et al. [3], was utilized in earlier experiments to accelerate inference by reducing the number of sampling steps while maintaining reasonable output fidelity.

**Algorithm 2** outlines the inference procedure.

Model evaluation was performed through qualitative visual inspection of outputs, using both in-distribution facial images and out-of-distribution general scenes. These tests assess the model's ability to generalize beyond the training distribution. The results highlight the effectiveness of the plug-and-play conditional diffusion approach in low-light enhancement scenarios, demonstrating its applicability to real-world deployments without retraining.

## 5. Experimental Results and Result Analysis

This section evaluates the performance of the trained illumination enhancement models under various training and inference configurations. Experiments were conducted to observe how different dataset sizes, training iterations, and sampling parameters affect restoration quality.

### 5.1 Qualitative Progression of Model Performance

Initial models trained with 1,000 image pairs for 2,000 iterations showed limited reconstruction fidelity. In particular, the restored images often exhibited color bias, with overemphasis on a single channel, most notably, excessive red saturation. These artifacts suggest that early models lacked sufficient exposure and training depth to generalize across varied lighting conditions and facial features. Figure 5 illustrates a representative failure case under this configuration, where the output image exhibits severe color dominance despite improved brightness.

#### 5.2 Improvement with Longer Training

Increasing the training to 10,000 steps (still using 1,000 image pairs) led to substantial improvements in both structure recovery and color balance. The model was able to eliminate the red channel saturation and reconstruct facial features with improved clarity. However, pixel-level noise remained visible thought the entire image, suggesting more timesteps are required during inference.

Inference for this configuration was tested across timestep counts of 25, 100, and 1000. The 500-step configuration was found to best balance denoising and detail preservation, offering cleaner results than 25 steps without the over smoothing observed at 1000 steps.

The final configuration used 10,000 paired images and trained for 20,000 steps, with inference conducted at 1000 timesteps. Despite the larger dataset and extended training, improvements were incremental. While general texture and lighting were well-preserved, the gains in noise reduction were small compared to the previous configuration. To further develop the model a larger variety of images could be used for training, using images of objects and scenery.

### 5.4 Figures of Experimental Outputs and Interpretation

To evaluate the model's performance across different data types and training configurations, three ground truth images were selected for testing:

- Image A: A brightly lit toy duck was used to assess the model's generalization beyond facial images, since the training set consisted solely of human faces.
- Image B: A male face that was not present in the training dataset, used to test generalization to unseen subjects.
- Image C: A face image that was included during training, serving as a reference for evaluating overfitting and reconstruction quality on familiar data.

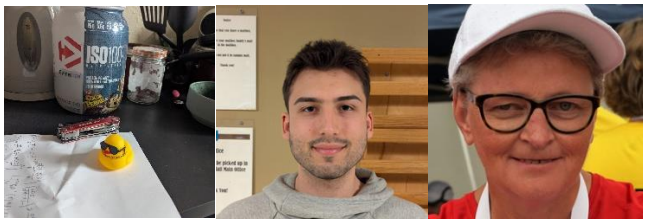

*Figure III:* Ground Truth (bright) Images for evaluation: Right: Image A; Middle: Image B; Left: Image C.

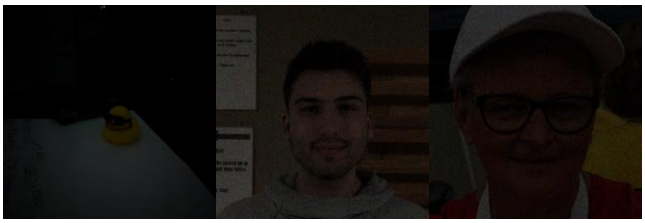

Figure IV: Dark Versions of bright images. Image A is retaken in dark, images B and C are synthetically darkened

Figure V: Output of model trained with 1,000 images, 2,000 steps, at inference with 25 timesteps, showcasing heavy color bias

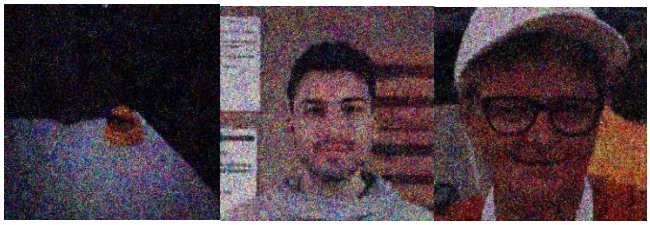

Figure VI: Output of model trained with 1,000 images, 10,000 steps, 25 timesteps, showcasing improved color balance

Figure VII: Output of final model trained with 10,000 images, 20,000 steps, 25 timesteps, showcasing improved brightness

Figure VIII: Output of final model trained with 10,000 images, 20,000 steps, 1000 timesteps. Showcasing removal of noise. The inference samples present with a small level of blur and distortion; however, information is well preserved. Noting readable text on the protein container behind the duck

The results demonstrate illumination and detail recovery; however, several artifacts remain. Some outputs exhibit residual blur, unnatural color shifts, color saturation, and exaggerated contrast or brightness levels, particularly in regions with complex textures or shadows. These issues indicate that the model has not fully converged and that its brightness capabilities can be further refined. Nonetheless, the outputs serve as a strong proof of concept for conditional diffusion-based illumination enhancement. With additional training iterations, incorporating non-facial images, and refined conditioning strategies, the model has the potential to achieve significantly higher fidelity in low-light image restoration.

## 6. Contribution

This project constitutes an extension of an existing idea. Specifically, Zhu et al.'s plug-and-play diffusion model framework for image restoration. While the original framework supports tasks such as denoising, deblurring, super-resolution, and image inpainting, this work expands its capabilities to include illumination enhancement. The implemented system enables the recovery of bright images from severely underexposed and dark inputs, introducing a new application domain for plug-and-play diffusion models.

Although this project builds upon publicly available code provided by Zhu et al., the contributions go beyond a reimplementation. Notable modifications include the design of a new training function to support image-pair conditioning (dark/bright) via channel expansion and the development of a custom inference script optimized for batch processing and deployment on consumer-grade hardware. These additions required substantial modifications to the original guided-diffusion pipeline by OpenAI [1], including adapting the architecture for conditional sampling with paired inputs.

*Zhu e al. code:* https://github.com/yuanzhi-zhu/DiffPIR.git

The project is accompanied by a suite of experiments detailed in Section 5, which investigate the effects of dataset size, training iterations, and timestep configurations on

model performance. Results show a consistent progression in output quality, validating the effectiveness of the proposed adaptations.

This work serves as proof of concept that image brightening is achievable using plug-and-play diffusion models. The ability to deploy pre-trained models without task-specific retraining, combined with the fast runtime performance of the inference script, highlights the potential of this approach for real-world applications such as night vision enhancement and CCTV footage restoration. Moreover, the positive trajectory observed through experimental iterations suggests that further research, dataset diversification, and training refinements could yield even higher-quality results.

With all said, this project successfully added the function of night vision to a general image restoration model and generated brightened samples from dark inputs.

**7. Conclusion and Future Directions**

The initial focus of this project until beyond Checkpoint 2 was to reimplement the deblurring pipeline from Zhu et al. However, after hitting a roadblock in that direction, the professor suggested pivoting to a new idea: applying the framework for Night Vision.

This project was grounded in Zhu's original framework, and while other research was not considered, the results were conclusive. The model works on something completely new, it improves with training, and it successfully extends plug-and-play functionality to a new restoration task.

Looking forward, preliminary steps could be studying state-of-the-art literature specifically focused on low-light image enhancement and night vision. Building architecture and training paradigms specifically tailored to that task, rather than retrofitting one designed for general image restoration could also be considered. Additionally, more extensive experimentation, especially regarding noise scheduling, loss functions, inference configurations, training data diversification could further improve performance. The goal of enhancing low light (near black) images was achieved, and a model to reconstruct bright images was developed.

## References


[1] A. Nichol and P. Dhariwal, "Improved Denoising Diffusion Probabilistic Models," arXiv preprint arXiv:2102.09672, 2021. [Online]. Available: https://github.com/openai/guided-diffusion

[2] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," Advances in Neural Information Processing Systems, vol. 33, pp. 6840–6851, 2020.

[3] J. Song, C. Meng, and S. Ermon, "Denoising diffusion implicit models," in International Conference on Learning Representations (ICLR), 2021. [Online]. Available: https://arxiv.org/abs/2010.02502

[4] T. Karras, S. Laine, and T. Aila, "A Style-Based Generator Architecture for Generative Adversarial Networks," Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR), pp. 4401–4410, 2019. [Online]. Available: https://github.com/NVlabs/ffhq-dataset

[5] Zhu, Y., Timofte, R., Van Gool, L., & Zhang, K. Denoising Diffusion Models for Plug-and-Play Image Restoration. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), pp. 1–10, 2023.

[6] N. Parikh, Stephen Boyd, et al. Proximal algorithms. Foundations and trends® in Optimization, 1(3):127–239, 2014.


## LLM Use Acknowledgement


This report was developed with assistance from a Large Language Model (LLM), specifically OpenAI's ChatGPT, used as a self-learning and support tool. The LLM was employed within the guidelines outlined in the course syllabus, serving to improve clarity, refine flow, and provide coding and conceptual guidance.

**LLM contributions included:**

**Technical support:** For example, clarifying concepts such as the purpose and origin of parameters like ddim25, and suggesting model training strategies with prompts like: "Give me an idea of where to start when training a new model, I want to do Night-Vision"

**Code assistance:** Providing scripts and debugging help, such as the transformation of image formats and preprocessing: "Create a script for transforming JPG images into PNG and then resize to 256x256 px."

**Writing refinement:** Suggestions were sought to enhance the structure and flow of the paper, using prompts such as: "While editing the paper, How do I organize the paragraphs and ideas for more consistent flow and clarity? Could someone get confused while reading this?"

**All content was authored by the student** and the final decisions regarding structure, content, and implementation were made independently. The LLM was not used to generate experimental results or to perform any unauthorized automation of the assignment.